IBM Fusion

Backup & Restore Lab Guide

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IBM Fusion: TechZone Quick Start

Introduction

IBM Technology Zone (TechZone) is the single destination for our go-to-market teams and IBM business partners ecosystem to access on-demand and live environments to learn, build, show, and share the value of IBM solutions. Additionally, they can extend our certified base images and further customize them for test, education, demonstration, and pilot activities.

This lab guide is a Quick Start for all Fusion labs on TechZone. The lab guide will explain how to find Fusion content, as well as how to reserve and use the Fusion related lab instances. Lab exercises and specific preparation steps are detailed in the respective lab guides by topic of interest.

This guide can be used to bring a TechZone instance to a general starting point. From this starting point you may continue to the relevant lab guide document or, if you wish, create your own specific configuration for a user defined test or demo.

All TechZone instances for Fusion are based on configurable OpenShift clusters with a virtual machine (VM) node infrastructure deployed on OpenShift Virtualization (OCP-Virt) technology. The underlying VMs have nested virtualization enabled so that they can host a second level VM for demonstrating technologies such as OCP-Virt and Hosted Control Planes (HCP aka, "Control Tower").

Product disclaimer

IBM Fusion is being developed and released in an agile manner. In addition to adding new capabilities, the interface is likely to change over time. Therefore, the screenshots used in this document and other lab guides may not always look exactly like what you see in the product. Depending timing, you can expect to encounter some of the following:

- Changes in the user interface (UI), such as the location of buttons or text in various fields
- Additional tabs or buttons

These differences should not affect how the lab exercises work. When changes are observed, have patience and explore.

You will need an IBM account and corresponding IBMid to gain access to IBM Technology Zone.

Getting help

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If you require assistance in interpreting any of the steps in this demo, please post your questions to the #storage_demo_feedback Slack channel for IBMers only. Business Partners can request help at the Partner Plus Support website.

If you have an issue with TechZone or want to open a support case go to the IBM Technology Zone Help. Alternatively, you can send an email to techzone.help@ibm.com. This contact information is also provided in the emails that you receive from IBM Technology Zone.

For help with the Fusion product see the Fusion documentation.

Prerequisites & getting started

You **must** be familiar with the Linux operating system, as well as know how to use the command line interface to execute basic Linux commands such as **kubectl**, **oc**, **exit**, **ls**, **mkdir**, **ssh**, **sudo**, and **curl** (to name few). Ideally, you should also have some experience working with the Red Hat OpenShift Graphical User Interface (GUI) and executing Red Hat OpenShift Container Platform and Kubernetes command line interface (CLI) commands. If you need experience using Linux, Kubernetes and Red Hat OpenShift Container Platform, it is highly recommended that you complete the following Red Hat courses *before* you attempt to work through the exercises in this lab.

Red Hat Courses:

- DO080 Containers Kubernetes and Red Hat OpenShift Technical Overview
- Red Hat OpenShift: Technical Overview

Note Red Hat Partner Connect Training access required.

When instructions are given to enter text, the text is usually shown in mono space font `this is an example`. This is to help indicate that you need to type or copy/paste all the text as is.

IBM Fusion on TechZone

IBM TechZone can be found at this URL https://techzone.ibm.com/. To access IBM TechZone you must have an IBM ID which you can obtain from your IBM representative as a client, Business Partner or employee.

Once you are logged in to IBM TechZone head over to the IBM Fusion Collection and explore the Environement and Resources.

Bookmark these pages for future use or mark them as Favorites by clicking the heart-shaped icon on the

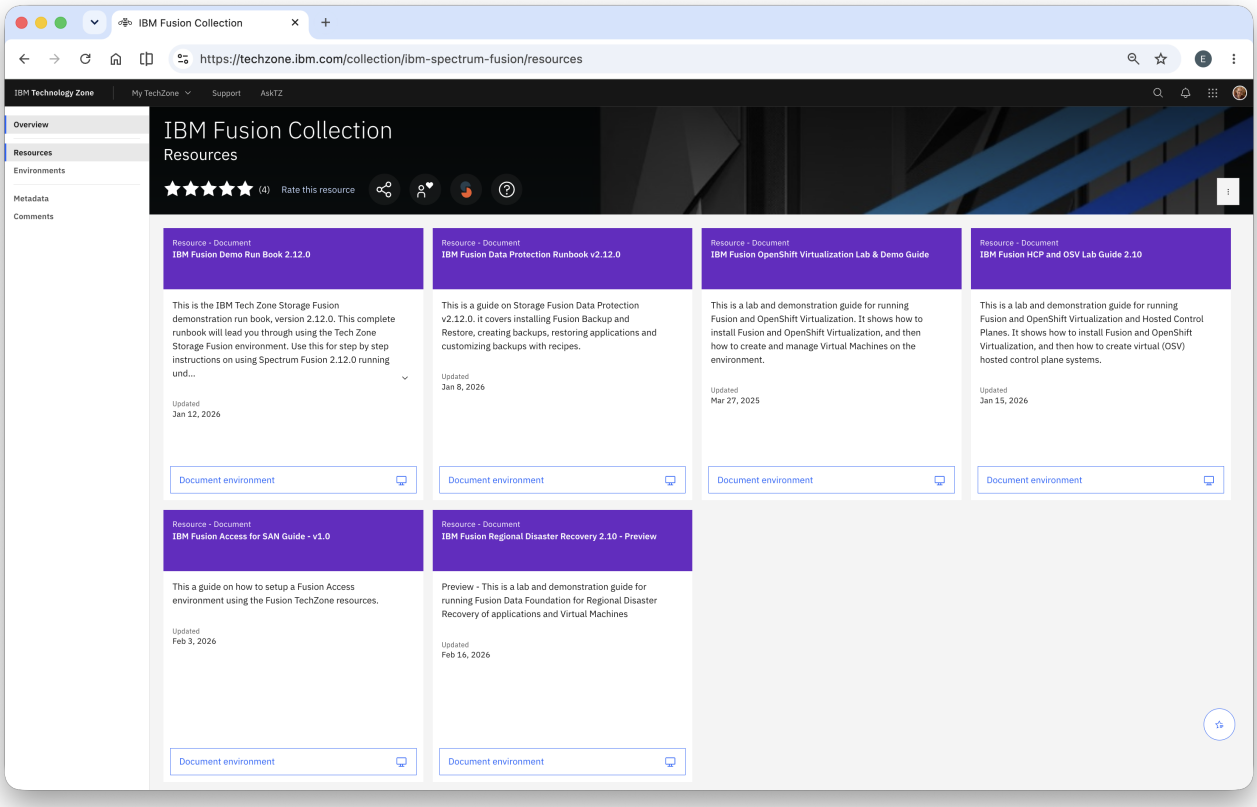
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page.

Resources

The resource page includes links to the available lab guides. These guides are available in PDF format and over time will be also available as on-line viewable material. Due to the rapid development cycles of some products, the guides may not always reflect the latest UI changes and you can expect to encounter some differences. These differences should not affect how the lab exercises work; be patience and explore to enhance your learning.

The lab guides are typically a step by step procedure to performing and learning a set of features and product capabilities. Feel free to explore and experiment!



Environments

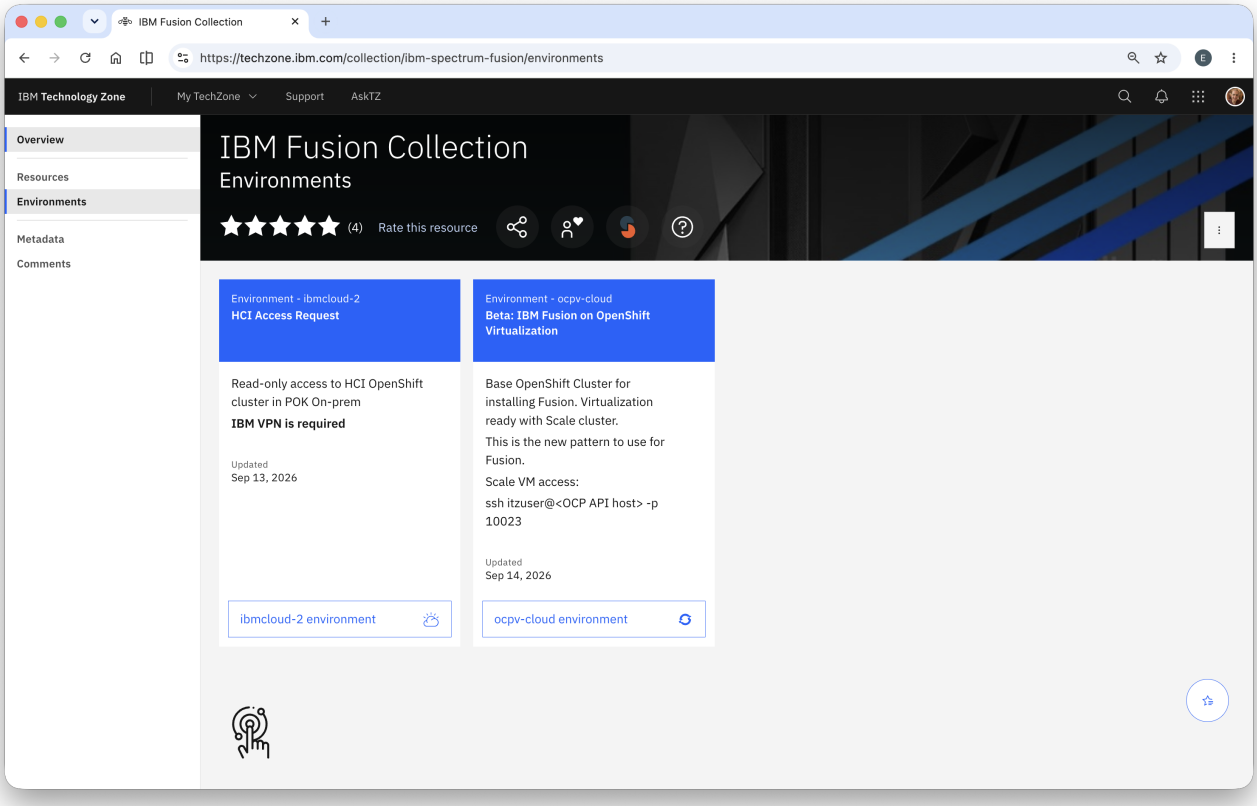
The environments page includes links to available lab resources that you can request for your project. Once your request is provisioned and ready, you will be able to use these resources for your own education, experimentation, and client demos.

The environment that is provisioned for you is an OpenShift (OCP) cluster consisting of your choice of

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components with some extra resources such as a Bastion node and an IBM Storage Scale node. You can use these for the guided labs or use them as a blank slate for your own explorations.

The next sections will walk you through a provisioning and basic use of an OCP cluster on TechZone.



Provisioning a lab

Begin your reservation process by logging in to TechZone and navigating the the Fusion Environments page.

The environments page lists the environment types you can provision and gain access to for your project.

Select the environment titled **Beta: IBM Fusion on OpenShift Virtualization** by clicking Reserve at the bottom of the tile. This will start a provisioning wizard that will walk you through the provisioning process:

1. General

Name your environment with a descriptive name that will help you identify the setup. This name will be visible to all authorized users, including team members, stakeholders, and clients.

Enter a free text description of the environment. Anyone you share this request with will see this

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description when getting access. Include relevant details to help them understand the purpose and expectations.

2. Business purpose

Describe the usage intent, including the context and purpose behind the request. This information helps the TechZone understand the business need and ensure the request is appropriately supported and prioritized.

By choosing a Purpose tile you can indicate the intended use. Each selection has a different reservation time schedule and different requirements for providing an ISC record reference. If you need to provide an ISC record reference simply copy and paste the 18-character Opportunity Number and make sure that the product list in the opportunity include the product name of the lab you are reserving; in our case Fusion.

3. Deployment

TechZone resource reservations rely on IBM Cloud. Choose a location closest to where the session will be hosted (if in person) to optimize performance. While this preference will be considered, deployment to the selected geography is subject to resource availability and cannot be guaranteed.

4. Scheduling

The scheduling page gives you the ability the set the preferred reservation date range within the Policy period determined by your intended use selection you made in the Business purpose page. Provide details on scheduling requirements and the planned dates for the intended request.

Important	Do not finalize dates with clients until the request is approved or until the resources are fully deployed and ready. Always reserve a few days in advance of any meetings or demos.

5. Configuration

Enter clear and detailed configuration choices for the environment request. You can choose to keep the defaults as displayed, or tailor the setup to your needs by clicking Customize and then modifying the environment attributes and then clicking Save.

6. Review and submit

Review the details on this pager before submitting your reservation request. This final step ensures that all required information is accurate and that you are aware of the responsibilities associated with the use of

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IBM TechZone resources.

You can go back and change reservation inputs by clicking the Edit links on the page or the Back button to step backwards in the wizard.

Any wizard step with a blue checkmark icon next to it is properly entered, and any red exclamation point icon should be revisited and corrected before submission.

Once you submit your request the system automatically start a background process to reserve the resources needed for configuring the OpenShift cluster and other resources.

You can track the reservation requests and their status on the My requests page. This page also has a display and sort filter that you can use the manage what you see on the page.

Tip Clear the filter or add the selection of Failed reservations so that you will be able to track reservations that failed to provision properly. This will prevent them from "disappearing" from your view.

You will receive emails from IBM Technology Zone <noreply@techzone.ibm.com> indicating the status of your reservation. The status may indicate a reservation failure or success, and over the duration of use you may also receive reminders for imminent expiration or confirmation of your initiated deletion.

If you receive and expiration notice and you have a date extension option, then follow up and extend your reservation in order to keep using it without disruption.

My reservation failed, what to do?

Reservations may fail from time to time. This depends of many factors including current workloads and available resources. Always reserve in advance of any meetings or demos! if a reservation fails redo the reservation entry using the reservation wizard and described in this section.

Managing a lab

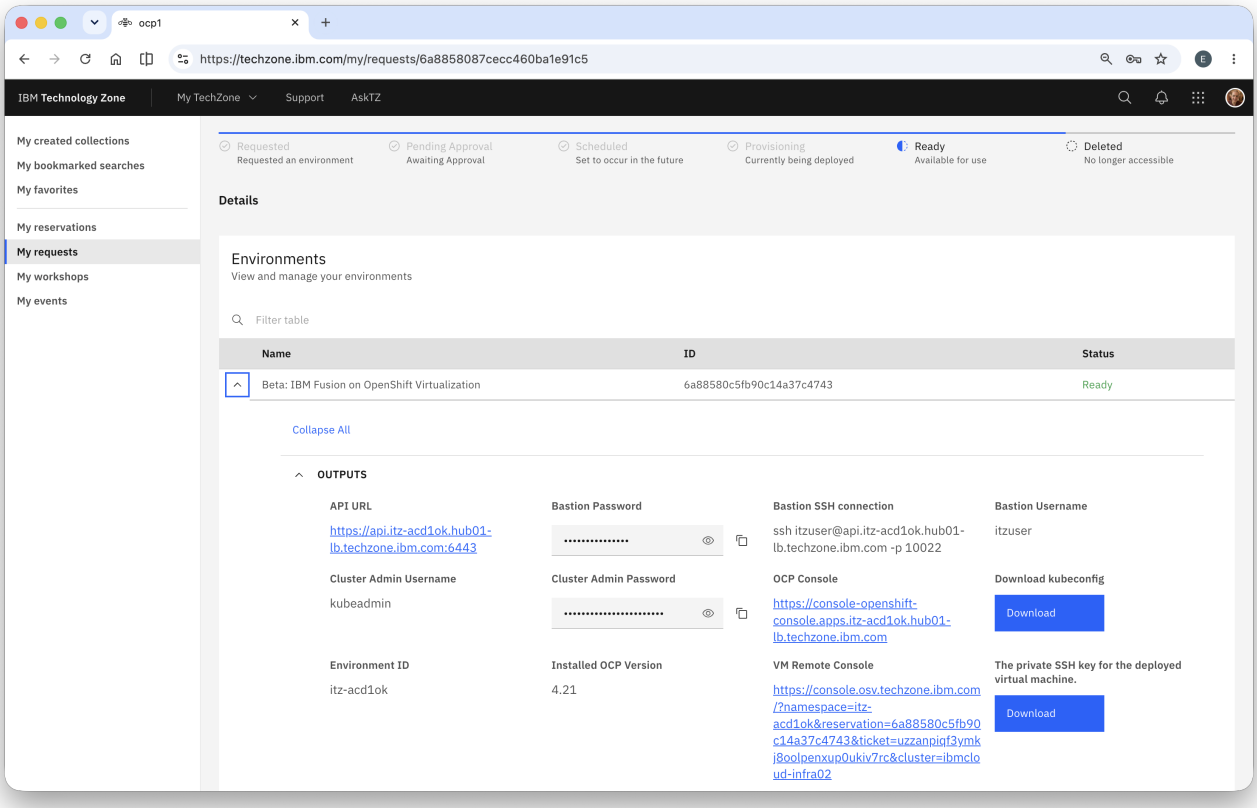
Once an environment reservation is successful you can start using it. If you are a first time then follow the steps below to get started.

Gather up some details

Start on the My requests page, locate the reservation you want to use, and make sure it is in the green "Reservation - Ready" state.

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Open the reservation page by clicking Open this reservation on the reservation tile (or the blue "eye" icon if you are in a list view instead of a tiled view.) Now expand the environment view by clicking the down-pointing arrow next to you environment name.



Before you continue we advise your to collects the various parameters of your environment such as passwords and access key file into a folder on your local desktop system. Create a folder which will be your work folder and collect the following artifacts into it:

- Download the kubeconfig file and save it
- Download the private SSH key and save it
- Create a text file and save the following parameters to it:
  - 24-character environment ID
  - Bastion user `itzuser` and the password from the hidden field
  - OCP cluster user `kubeadmin` and the password from the hidden field

For detailed instructions on connecting to the Web Console, SSH-ing to the Bastion, installing `oc` locally, and VM IP address references, see Accessing the Environment.

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Accessing Your TechZone Lab Environment

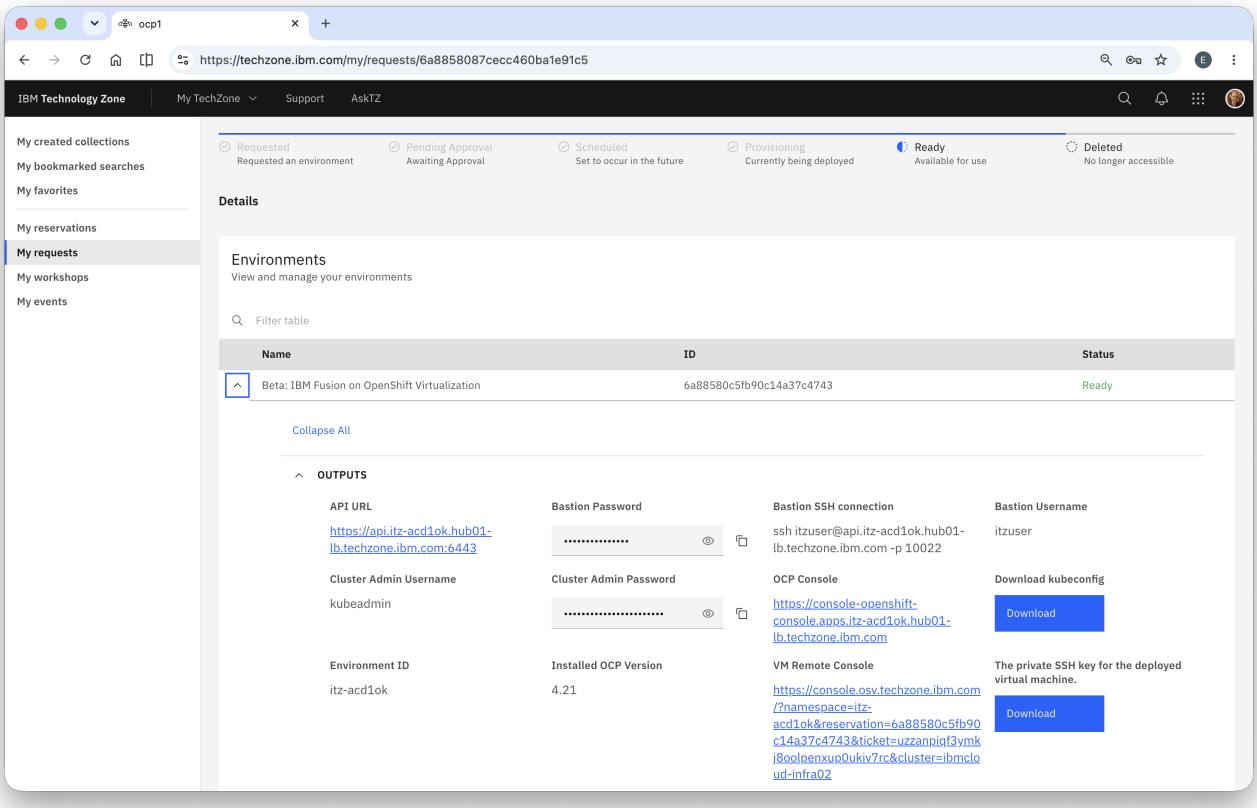
Introduction

Once your TechZone reservation is in the **Ready** state, you have three ways to interact with your OpenShift cluster. Most learners use a combination — the Web Console for UI-driven tasks and either the Bastion or a local `oc` install for CLI work.

Note If you have not yet provisioned a TechZone environment, complete the TechZone Quick Start first.

Gathering Your Credentials

All endpoints and credentials are found in the **Outputs** section of your reservation page. Navigate to My requests, open your reservation, and expand the environment row to reveal the Outputs panel shown below.



Before proceeding, collect the following into a working folder on your local machine:

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- Download the **kubeconfig** file — saves as `conf_kubeconfig_download.conf`
- Download the **private SSH key** — saves as `vm_ssh_key.vm` (check your Downloads folder)
- Note or copy the following values from the hidden fields (click the eye icon to reveal):
  - **Cluster Admin Password** (username is `kubeadmin`)
  - **Bastion Password** (username is `itzuser` — used for `sudo` on the Bastion, not SSH login)
- Note the **Bastion SSH connection** command — it is pre-built with the correct hostname and port
- Note the **API URL** — used for `oc login`

The following table describes every field in the Outputs panel:

Field	Description
API URL	OpenShift API endpoint used for `oc login` and kubeconfig
OCP Console	OpenShift Web Console URL — open directly in your browser
Cluster Admin Username	`kubeadmin`
Cluster Admin Password	Hidden — click the eye or copy icon
Bastion SSH connection	Pre-built SSH command — copy and paste directly into your terminal
Bastion Username	`itzuser`
Bastion Password	Hidden — required only for `sudo` on the Bastion, not for SSH login
Download kubeconfig	Download — saves as `conf_kubeconfig_download.conf`
The private SSH key	Download — required to authenticate SSH to the Bastion
VM Remote Console	Browser-based VNC access to the underlying cluster VMs
Environment ID	Your unique 24-character environment identifier
Installed OCP Version	The OpenShift version running in your cluster

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Important The Bastion host uses **key-based SSH authentication only** on port 10022. Password authentication is disabled on that port. You must download the private SSH key before attempting to connect.

Choosing an Access Method

Method	Best for	Requires
OpenShift Web Console	UI-driven tasks, monitoring, Software Catalog	Browser only
Bastion host (SSH)	CLI tasks — `oc` is pre-installed, no local setup needed	SSH client + private key
Local `oc` on your laptop	CLI tasks without connecting to the Bastion	`oc` installed + kubeconfig

Option 1: OpenShift Web Console

The OpenShift Web Console runs entirely in your browser — no software installation required.

1. From your TechZone reservation page, copy the **OCP Console** URL.
2. Open it in your browser.
3. On the login page, select the **kube:admin** identity provider.

   Tip Select **kube:admin** rather than IBMid to ensure you log in with full `cluster-admin` privileges.

4. Enter `kubeadmin` as the username and paste the **Cluster Admin Password** from your reservation outputs.
5. Upon successful authentication the OpenShift Administrator Dashboard is displayed.

Tip When your browser offers to save the password, accept — you will be logging back in frequently throughout the labs.

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Option 2: Bastion Host via SSH

The Bastion host has `oc`, `kubectl`, `helm`, and `curl` pre-installed. Connecting via SSH is the quickest way to run CLI commands without any local setup.

Step 1: Download the private SSH key

On your TechZone reservation page, click Download next to **The private SSH key for the deployed virtual machine**. The file saves as `vm_ssh_key.vm` (check your Downloads folder).

Step 2: Set key permissions

SSH requires the private key file to be readable only by your user.

macOS / Linux:

```
chmod 400 ~/Downloads/vm_ssh_key.vm
```

Windows (PowerShell):

```
icacls "$env:USERPROFILE\Downloads\vm_ssh_key.vm" /inheritance:r /grant:r
"$env:USERNAME:(R)"
```

Step 3: Connect via SSH

Copy the **Bastion SSH connection** command from your reservation page — it is pre-populated with the correct hostname and port. Add the `-i` flag pointing to your downloaded key:

macOS / Linux:

```
ssh -i ~/Downloads/vm_ssh_key.vm itzuser@api.<env-id>.techzone.ibm.com -p 10022
```

Windows (PowerShell — OpenSSH is built in to Windows 10 and 11):

```
ssh -i "$env:USERPROFILE\Downloads\vm_ssh_key.vm" itzuser@api.<env-id>.techzone.ibm.com -p
10022
```

Note If you are on an older Windows version without built-in OpenSSH, use PuTTY. Convert `vm_ssh_key.vm` to PuTTY format using **PuTTYgen** (Load → Save private key), then connect via PuTTY on port 10022.

Step 4: Log in to OpenShift from the Bastion

Once connected to the Bastion, log in to your OpenShift cluster using the **API URL** and **Cluster Admin**

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Password from your reservation:

```
oc login -u kubeadmin -p <cluster-admin-password> https://api.<env-
id>.techzone.ibm.com:6443
```

Verify connectivity:

```
oc whoami
oc get nodes
```

`oc whoami` should return `kubeadmin` and `oc get nodes` should list all cluster nodes in `Ready` state.

Option 3: Local `oc` CLI on Your Laptop

If you prefer to run `oc` commands directly from your laptop without SSH-ing to the Bastion, install the `oc` CLI and use the downloaded kubeconfig.

Step 1: Install the `oc` CLI

macOS:

```
# Using Homebrew
brew install openshift-cli

# Verify
oc version
```

Alternatively, download the binary from the OCP Web Console: click the Help icon (?) in the top-right masthead and select Help › Command Line Tools, then download the macOS binary and move it to `/usr/local/bin/`.

Linux:

```
# Download from the OCP Web Console (Help > Command Line Tools), or use the mirror:
curl -LO https://mirror.openshift.com/pub/openshift-v4/clients/ocp/stable/openshift-
client-linux.tar.gz
tar -xvf openshift-client-linux.tar.gz
sudo mv oc /usr/local/bin/
oc version
```

Windows:

Download the `oc` binary from the OCP Web Console (Help icon → Help › Command Line Tools → **Windows**). Extract the zip and move `oc.exe` to a folder on your `PATH` (e.g. `C:\tools\`).

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Alternatively, install via Chocolatey:

```
choco install openshift-cli
oc version
```

Step 2: Download the kubeconfig

On your TechZone reservation page, click Download next to **Download kubeconfig**. The file saves as `conf_kubeconfig_download.conf`.

Step 3: Connect using the kubeconfig

macOS / Linux:

```
export KUBECONFIG=~/Downloads/conf_kubeconfig_download.conf
oc whoami
oc get nodes
```

Windows (PowerShell):

```
$env:KUBECONFIG = "$env:USERPROFILE\Downloads\conf_kubeconfig_download.conf"
oc whoami
oc get nodes
```

A successful connection returns `system:admin` from `oc whoami`.

Tip

To make the kubeconfig permanent across terminal sessions, merge it into your default `~/.kube/config`:

```
# macOS / Linux
KUBECONFIG=~/.kube/config:~/Downloads/conf_kubeconfig_download.conf \
  oc config view --flatten > /tmp/merged && mv /tmp/merged ~/.kube/config
```

SSH to IBM Storage Scale node

Option 1: Bastion node

To SSH from the Bastion directly into the IBM Storage Scale node:

```
ssh itzuser@10.10.10.202 □
```

□ Verify this IP against the VM Remote Console page before use.

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Option 2: Directly from your workstation

macOS / Linux:

```
ssh -i ~/Downloads/vm_ssh_key.vm itzuser@api.<env-id>.techzone.ibm.com -p 10023
```

Windows (PowerShell — OpenSSH is built in to Windows 10 and 11):

```
ssh -i "$env:USERPROFILE\Downloads\vm_ssh_key.vm" itzuser@api.<env-id>.techzone.ibm.com -p 10023
```

Note If you are on an older Windows version without built-in OpenSSH, use PuTTY. Convert `vm_ssh_key.vm` to PuTTY format using **PuTTYgen** (Load → Save private key), then connect via PuTTY on port 10023.

VM IP Address Reference

The following IP addresses are assigned to VMs in the standard TechZone Fusion environment. Your configuration may include additional worker nodes — treat this as a representative reference:

VM	IP Address
`itz-*-bastion-1`	`10.10.10.201`
`itz-*-scale-*`	`10.10.10.202`
`itz-*-master-1`	`10.10.10.11`
`itz-*-master-2`	`10.10.10.12`
`itz-*-master-3`	`10.10.10.13`
`itz-*-infra-1`	`10.10.10.2`
`itz-*-infra-2`	`10.10.10.3`
`itz-*-infra-3`	`10.10.10.4`
`itz-*-worker-1`	`10.10.10.21`

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VM	IP Address
`itz-*-worker-2`	`10.10.10.22`
`itz-*-worker-3`	`10.10.10.23`

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IBM Fusion Operator Installation & Console Overview

Introduction

Although your Red Hat OpenShift cluster is provisioned on IBM Technology Zone, IBM Fusion has not yet been deployed. In this guide, you will install the IBM Fusion Operator from the Software Catalog, accept the product license, and explore the core administration interface.

Note This lab requires an active IBM Fusion reservation on IBM Technology Zone (TechZone). For step-by-step guidance on creating an IBMid, reserving an environment, and retrieving your cluster credentials, see the TechZone Quick Start Guide.

Installing the IBM Fusion Operator

The IBM Operator Catalog and entitlement secrets are pre-configured on your TechZone OpenShift cluster.

Install via OpenShift Web Console

1. Log in to the OpenShift Web Console as `kubeadmin`.
2. In the left navigation menu, go to Ecosystem › Software Catalog.
3. In the search field, type `Fusion` and select the **IBM Storage Fusion** operator tile.
4. Click Install.
5. Leave all installation options as default:
   - **Update channel:** Stable channel (e.g., `v2.x`)
   - **Installation mode:** All namespaces on the cluster (default)
   - **Installed Namespace:** `ibm-spectrum-fusion-ns` (or Operator-recommended namespace)
   - **Approval Strategy:** Automatic
6. Click Install at the bottom of the form.
7. Wait until the Operator status shows **Succeeded** / **Installed operator - ready for use**.

Optional: Install via OpenShift CLI (`oc`)

You can also verify or initiate operator installation from the Bastion terminal:

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```
oc get csv -n ibm-spectrum-fusion-ns
```

Connecting to the IBM Fusion Management Console

IBM Fusion is container-native and integrates directly into the OpenShift console.

1. From the top masthead of the OpenShift Web Console, click the **Application Menu** icon (the 9-square grid).
2. Click **IBM Storage Fusion** from the dropdown menu. A new browser tab opens to the Fusion interface.
3. If prompted with the OpenShift login page, select **kube:admin** and enter your cluster credentials.
4. On first launch, review and select **I have read and accept the license agreement**, then click Continue.

> **Note** Accepting the license initializes the core `SpectrumFusion` Custom Resource (CR) in the `ibm-spectrum-fusion-ns` namespace.

5. You are redirected to the **Welcome to IBM Fusion** dashboard.

Navigating the IBM Fusion Dashboard

The IBM Fusion interface provides a unified view of data management services:

- **Services:** View installed services and install optional capabilities (Fusion Data Foundation, Backup & Restore, Data Cataloging).
- **Storage:** Manage local storage clusters, StorageClasses, and remote IBM Storage Scale file systems.
- **Backup & Restore:** Configure backup locations (S3), retention policies, and application protection.
- **Events:** View Kubernetes-native events with severity filtering (Info, Warning, Critical).
- **Settings:** Review cluster networking, Call Home diagnostics, and cluster interconnect status.
- **Help & Support:** Access the question-mark icon in the masthead to download support logs or link to official IBM documentation.

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Configuring Fusion Data Foundation (FDF)

Introduction

In this guide you will enable the Data Foundation service in the IBM Fusion Console and configure the local Ceph storage cluster using the OpenShift Web Console wizard.

Note This lab requires an active IBM Fusion reservation on IBM Technology Zone (TechZone). For step-by-step guidance on creating an IBMid, reserving an environment, and retrieving your cluster credentials, see the TechZone Quick Start Guide.

Enabling the Data Foundation Service

Note This guide covers the **Local** configuration mode, the standard mode for single-cluster labs. For an overview of all FDF configuration modes (Local, Dynamic, External, and Provider), see Persistent Storage: Architecture & Concepts.

1. In the IBM Fusion Console, navigate to **Services** in the left navigation menu.
2. Locate the **Data Foundation** service tile, click the tile, and then click Install.
3. On the **Install service** screen, select the **Local** tile and click Install.

   Note Selecting **Local** automatically installs the Local Storage Operator (LSO) — no separate LSO installation step is required.

4. Wait for the service status to show **Healthy**.

Configuring the Ceph Storage Cluster

Once the Data Foundation service is enabled, configure the local Ceph storage cluster through the OpenShift Web Console wizard launched from Fusion:

Step 1: Launch Storage Cluster Creation

1. In the IBM Fusion Console, navigate to Storage › Data Foundation (or **Local storage**).
2. Click Get Started (or Configure Data Foundation / Create Storage Cluster).

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3. This redirects to the **Create StorageSystem** wizard in the OpenShift Web Console.

Step 2: Backing Storage

1. On the **Backing storage** page, select **Create a new StorageClass using local storage devices**.
2. Click Next.

Step 3: Advanced Settings

1. On the **Advanced Settings** page, configure the default StorageClass options:
   - Leave **Use Ceph RBD as the default StorageClass** checked. This sets Ceph RBD as the default StorageClass, eliminating the need to manually annotate a StorageClass after installation.
   - If this cluster will host virtual machines, also check **Set default StorageClass for virtualization**. This marks the RBD virtualization StorageClass as the default for KubeVirt VM disks after installation.
2. Leave all other options at their defaults and click Next.

Note Both default StorageClass options are configured here in the wizard. No additional manual annotation steps are required after installation.

Step 4: Create Local Volume Set

1. Enter `ibm-spectrum-fusion` as the **LocalVolumeSet name**.
2. Select **Disks on selected nodes** and choose the dedicated infra nodes (`itz-*-infra-1`, `itz-*-infra-2`, `itz-*-infra-3`).
3. Set **Disk Type** to **SSD/NVMe**.
4. Click the `^` to reveal the **Advanced** section, and enter "100" in the minimum drive size field; this will exclude the 50GiB drives.
5. Click Next, then click Yes to confirm creation of the LocalVolumeSet.

Step 5: Capacity and Nodes

1. Verify the **Available raw capacity** populated from the discovered disks.
2. Select a **resource profile** — **Balanced** is recommended for lab environments.
3. Click Next.

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Step 6: Security and Network

1. Keep the default encryption settings unless KMS integration (Vault or Thales CipherTrust) is required.
2. Keep the default network option (**Default (OVN)**) unless using Multus.
3. Click Next.

Note If you are configuraing storage for Hosted Control Plane (HCP) then select the "Host" radio button.

Step 7: Data Protection

1. If this cluster will be configured for Regional Disaster Recovery, select **Prepare cluster for disaster recovery (Regional-DR only)**.
2. Otherwise leave unchecked and click Next.

Step 8: Review and Create

1. Review the configuration summary. Click Back to correct any settings.
2. Click Create StorageSystem.

Note Storage cluster initialization takes approximately 10–15 minutes. Wait until the cluster status reports **Ready / Healthy** under Storage › Data Foundation in the OpenShift console before proceeding.

Step 9: Verify StorageClasses and Ceph Cluster Health

From the OpenShift Web Console:

1. Navigate to Storage › StorageClasses and confirm the standard FDF storage classes are listed.
2. Navigate to Storage › Data Foundation and confirm the storage cluster status shows **Ready / Healthy**.
3. Navigate to Ecosystem › Installed Operators › OpenShift Data Foundation and confirm the operator status shows **Succeeded**.

From the CLI:

```
# Check provisioned StorageClasses
oc get sc

# Verify Ceph cluster status
```


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```
oc get storagecluster -n openshift-storage

# Confirm all ODF pods are Running or Completed
oc get pods -n openshift-storage
```

The following standard StorageClasses should now be available:

- `ocs-storagecluster-ceph-rbd` (Block storage for containers & VMs)
- `ocs-storagecluster-cephfs` (Shared file storage / RWX)
- `openshift-storage.noobaa.io` (Object Bucket Claims / S3)

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Installing the Backup & Restore Service

Introduction

The IBM Fusion Backup & Restore service is an optional capability that must be explicitly installed from the Fusion Services catalog. This page walks through applying a required hotfix, installing the service from the Fusion console, and verifying that all service pods are healthy before you proceed to configure backup locations and policies.

Note This lab requires an active IBM Fusion reservation on IBM Technology Zone (TechZone). For step-by-step guidance on creating an IBMid, reserving an environment, and retrieving your cluster credentials, see the TechZone Quick Start Guide.

Required Hotfix

Warning

STOP — Apply This Hotfix Before Proceeding

All IBM Storage Fusion Backup & Restore releases are affected.

IBM has identified a critical issue: the MinIO container image previously hosted on `quay.io/minio` has been **permanently removed** from the public Quay registry. Without this fix applied first, the Backup & Restore service installation will fail, the Transaction Manager pod will not start, and all backup and restore jobs will be blocked with an image-pull authorization error on `guardian-minio`.

Do not proceed to the installation steps below until this hotfix has been applied and the MCP rollout is complete.

Option A — Apply via OpenShift Web Console:

1. Navigate to Administration › CustomResourceDefinitions.
2. In the search field, type `ImageDigestMirrorSet` and open the CRD.
3. Click the **Instances** tab, then click Create ImageDigestMirrorSet.
4. Replace the default YAML editor content with the manifest below, then click Create.

```
apiVersion: config.openshift.io/v1
kind: ImageDigestMirrorSet
```


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```
metadata:
  name: fbr-minio-idms
spec:
  imageDigestMirrors:
  - mirrors:
    - icr.io/fusion-open/minio
    source: quay.io/minio
```

Option B — Apply via CLI:

```
cat <<'EOF' | oc create -f -
apiVersion: config.openshift.io/v1
kind: ImageDigestMirrorSet
metadata:
  name: fbr-minio-idms
spec:
  imageDigestMirrors:
  - mirrors:
    - icr.io/fusion-open/minio
    source: quay.io/minio
EOF
```

Verify the MCP rollout is complete before continuing:

```
oc get mcp
```

The rollout triggers automatically and does **not** reboot any nodes. Wait until all `MachineConfigPool` entries report `UPDATED=True` and `DEGRADED=False` before proceeding.

For the full IBM Support advisory, see: MinIO Image Removal from Quay Registry — Advisory 7289585.

Installing the Backup & Restore Service

Install via IBM Fusion Console

1. Open the IBM Fusion console (Application Menu › IBM Storage Fusion from the OpenShift masthead).
2. In the left navigation, click **Services**.
3. Locate the **Backup & Restore** tile and click Install.

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4. Review the installation options — leave all defaults — and click Install to confirm.
5. The tile status changes to **Installing**. Wait until it shows **Installed**.

> **Note** Installation typically takes 5–10 minutes. You can monitor pod startup in real time with the CLI command in the next section.

Verify Installation via CLI

Monitor pod readiness in the `ibm-spectrum-fusion-ns` namespace:

```
oc get pods -n ibm-spectrum-fusion-ns -w | grep -E 'guardian|minio|transaction'
```

Wait until all `guardian-`, `guardian-minio-`, and `transaction-manager-*` pods show `Running` with all containers ready (e.g. `1/1` or `2/2`).

To get a quick summary of any pods that are not yet running:

```
oc get pods -n ibm-spectrum-fusion-ns --field-selector=status.phase!=Running | grep -E
'guardian|minio|transaction'
```

An empty result confirms all relevant pods are running.

Confirming Topology Connectivity

After installation, Backup & Restore registers the local cluster as a **Hub** node in its topology.

Verify via IBM Fusion Console

1. In the Fusion console, navigate to Backup & restore › Topology.
2. Confirm the cluster tile shows a status of **Connected**.

Verify via CLI

```
oc get spectrumfusionbackup -n ibm-spectrum-fusion-ns
```

The `STATUS` column should show `Ready`.

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IBM Fusion Backup & Restore: Architecture & Overview

Introduction

IBM Fusion Backup & Restore provides enterprise data protection and disaster recovery for stateful, containerized applications running on Red Hat OpenShift. It protects not only persistent data volumes (PVCs) but also all associated Kubernetes metadata — Deployments, StatefulSets, ConfigMaps, Secrets, Routes, and Services.

About this Lab

This lab is divided into two main sections:

1. **Configure and use Backup & Restore on the Hub cluster**
   - Define backup locations and policies.
   - Deploy a sample application and assign backup policies.
   - Restore an application to a new namespace.
   - Monitor backup and restore job status.
   - Create a Fusion Recipe for application-consistent backups (MongoDB/Pacman).
   - Configure Backup Service Protection.
2. **Hub-and-Spoke: back up on one cluster, restore to another**
   - Install the Backup & Restore agent on a second TechZone cluster.
   - Create a shared S3 object store accessible from both clusters.
   - Back up an application on the Hub and restore it to the Spoke.

Important Two TechZone IBM Fusion environments are required for this lab. Reserve both before you begin — it helps to provision them at the same time. The cluster with the full Backup & Restore service installed is the **Hub**. The second cluster with only the agent installed is the **Spoke**.

Note This lab requires an active IBM Fusion reservation on IBM Technology Zone (TechZone). For step-by-step guidance on creating an IBMid, reserving an environment, and retrieving your cluster credentials, see the TechZone Quick Start Guide.

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Key Capabilities

- **Application-Centric Protection:** Protects PVCs and all associated Kubernetes resources together as a single application unit.
- **Flexible Storage Targets:** Back up to local in-place snapshots (Ceph/FDF CSI snapshots) or external S3-compatible object storage (NooBaa/MCG, IBM COS, AWS S3).
- **Multi-Cluster Hub-and-Spoke:** A single Fusion Hub cluster orchestrates backups across distributed Spoke clusters via a lightweight agent.
- **Application Consistency via Recipes:** Execute pre- and post-backup hooks to quiesce databases before snapshots are taken.
- **Service Protection:** Back up the Backup & Restore control plane itself to S3 so the service can be recovered after a cluster failure.

Hub vs. Spoke Architecture

IBM Fusion Backup & Restore operates across two cluster personas:

Hub Cluster

Runs the full Backup & Restore service — controller, inventory, scheduler, policy manager, and UI. Also runs a local agent for protecting applications on the Hub itself.

Spoke Cluster

Runs only the lightweight Backup & Restore agent. The agent receives scheduled tasks from the Hub and executes local snapshotting and data movement. All scheduling, retention, policy handling, and location management remain on the Hub.

The Backup & restore › Topology page shows the Hub and all connected Spoke clusters. Use it to verify connectivity before starting lab exercises.

Software Versions

The following software versions were used to develop and validate the Backup & Restore lab:

Component	Version
Red Hat OpenShift Container Platform	4.21.33

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Component	Version
IBM Fusion	2.14
Fusion Data Foundation	4.21.12
Fusion Backup & Restore	2.14

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Configuring Backup Locations & Policies

Introduction

Before protecting application workloads, you must configure a **Backup Location** (where data copies are stored) and **Backup Policies** (schedules and retention rules). In this lab you will create one S3 object storage location backed by the local FDF NooBaa/MCG bucket, then create two policies — a daily in-place snapshot and a weekly object storage backup.

Note This lab requires an active IBM Fusion reservation on IBM Technology Zone (TechZone). For step-by-step guidance on creating an IBMid, reserving an environment, and retrieving your cluster credentials, see the TechZone Quick Start Guide.

Note Navigate to Backup & restore › Topology and confirm the cluster shows as **Connected** before proceeding.

Create the S3 Backup Location

Step 1: Create an Object Bucket Claim

From the OpenShift Web Console:

1. Navigate to Storage › Object Storage.
2. Click the **Object Bucket Claims** tab.
3. Click Create ObjectBucketClaim.
4. Set:
   - **ObjectBucketClaim name:** `backup-bucket-claim`
   - **StorageClass:** `openshift-storage.noobaa.io`
   - **BucketClass:** `noobaa-default-bucket-class` (default)
5. Click Create.
6. Once the status shows **Bound**, scroll down to **Object Bucket Claim Data** and click Reveal Values.
7. Note the **Bucket Name**, **Access Key**, and **Secret Key** — you will need them in Step 3.

Verify the ObjectBucketClaim (OBC) is Bound and retrieve the bucket credentials from the CLI:

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Confirm the OBC is Bound

```
oc get obc backup-bucket-claim
```

Retrieve the bucket name

```
oc get obc backup-bucket-claim -o jsonpath='{.spec.bucketName}'
```

Retrieve the access credentials

```
oc get secret backup-bucket-claim -o jsonpath='{.data.AWS_ACCESS_KEY_ID}' | base64 -d
oc get secret backup-bucket-claim -o jsonpath='{.data.AWS_SECRET_ACCESS_KEY}' | base64 -d
```

Step 2: Retrieve the External S3 Route

The internal endpoint shown on the OBC details page is not accessible outside the cluster. Retrieve the external route instead.

From the CLI:

```
oc get route s3 -n openshift-storage -o jsonpath='{.spec.host}'
```

From the OpenShift Web Console:

Navigate to Networking › Routes, change the project to `openshift-storage`, and copy the **Location** URL for the `s3` route.

Step 3: Register the Location in IBM Fusion

1. In the IBM Fusion Console, navigate to Backup & restore › Locations.
2. Click Add location +.
3. Enter `object1` in the **Location name** field.
4. Click the **MCG/NooBaa** tile, then click Next.
5. Enter the connection details:
    - **Endpoint:** `https://<s3-route-host>` (from Step 2)
    - **Bucket Name:** `<bucket-name>` (from Step 1)
    - **Access Key:** `<access-key>` (from Step 1)
    - **Secret Key:** `<secret-key>` (from Step 1)
6. Click Add. Wait for the location tile to show **Connected**.

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Create Backup Policies

Policy 1: Daily In-Place Snapshot

In-place snapshots use local CSI volume snapshots and do not require an external object storage location.

1. Navigate to Backup & restore › Policies.
2. Click Add policy +.
3. Set:
   - **Policy name:** `daily-snapshot`
   - **Frequency:** `Daily`
   - **Backup location:** Select the **In place snapshot** tile.
4. Specify a **Time window** and click Create policy.

Policy 2: Weekly Object Storage Backup

1. Click Add policy +.
2. Set:
   - **Policy name:** `weekly-backup`
   - **Frequency:** `Weekly`
   - **Schedule:** `Sunday`
3. Select the **Object Storage** tile, then select the `object1` location.
4. Specify a **Time window** and click Create policy.

The Policies summary page shows each policy's schedule, storage location, and which applications are assigned to it.

Verify policies were created from the CLI:

```
oc get backuppolicy -n ibm-spectrum-fusion-ns
```

Both `daily-snapshot` and `weekly-backup` should appear in the output.

Tip Always use an Object Storage backup location for any backup you may need to restore across clusters. In-place snapshots are local to the cluster and cannot be used for cross-cluster restore.

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Protecting Workloads

Introduction

In this exercise you will deploy a stateful file browser application, protect it with both backup policies created in the previous lab, trigger an initial backup, monitor its progress, and restore the application to a new namespace.

Note This lab requires an active IBM Fusion reservation on IBM Technology Zone (TechZone). For step-by-step guidance on creating an IBMid, reserving an environment, and retrieving your cluster credentials, see the TechZone Quick Start Guide.

Deploy the Filebrowser Application

Create the Project

From the OpenShift Web Console:

1. Navigate to Home › Projects. Click Create Project.
2. Enter `filebrowser` in the **Name** field and click Create.

From the CLI:

```
oc new-project filebrowser
```

Import the Application YAML

1. Verify the active project is `filebrowser`.
2. Click the Quick create icon (+) in the masthead and select **Import YAML**.
3. Paste the following YAML and click Create:

```
kind: PersistentVolumeClaim
apiVersion: v1
metadata:
  labels:
    app: filebrowser
    app.kubernetes.io/component: filebrowser
    app.kubernetes.io/instance: filebrowser
```

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```
    app.kubernetes.io/name: filebrowser
    app.kubernetes.io/part-of: filebrowser-app
  name: files
spec:
  storageClassName: ocs-storagecluster-ceph-rbd
  accessModes:
    - ReadWriteOnce
  resources:
    requests:
      storage: 10Gi

---
kind: PersistentVolumeClaim
apiVersion: v1
metadata:
  labels:
    app: filebrowser
    app.kubernetes.io/component: filebrowser
    app.kubernetes.io/instance: filebrowser
    app.kubernetes.io/name: filebrowser
    app.kubernetes.io/part-of: filebrowser-app
  name: filebrowser-config
spec:
  storageClassName: ocs-storagecluster-ceph-rbd
  accessModes:
    - ReadWriteOnce
  resources:
    requests:
      storage: 1Gi

---
kind: ConfigMap
apiVersion: v1
metadata:
  name: my-filebrowser-config
  labels:
    app: filebrowser
    app.kubernetes.io/component: filebrowser
    app.kubernetes.io/instance: filebrowser
    app.kubernetes.io/name: filebrowser
    app.kubernetes.io/part-of: filebrowser-app
data:
```


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```
  .filebrowser.json: |
    {
      "port": 8080,
      "baseURL": "",
      "address": "",
      "log": "stdout",
      "database": "/config/database.db",
      "root": "/files"
    }

---
kind: Deployment
apiVersion: apps/v1
metadata:
  name: filebrowser
  labels:
    app: filebrowser
    deployment: filebrowser
spec:
  replicas: 1
  selector:
    matchLabels:
      app: filebrowser
  template:
    metadata:
      labels:
        app: filebrowser
        app.kubernetes.io/component: filebrowser
        app.kubernetes.io/instance: filebrowser
        app.kubernetes.io/name: filebrowser
        app.kubernetes.io/part-of: filebrowser-app
    spec:
      restartPolicy: Always
      serviceAccountName: default
      schedulerName: default-scheduler
      enableServiceLinks: true
      terminationGracePeriodSeconds: 30
      securityContext: {}
      containers:
        - resources:
            requests:
              cpu: "100m"
```

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```
        memory: "128Mi"
      limits:
        cpu: "500m"
        memory: "512Mi"
    readinessProbe:
      httpGet:
        path: /health
        port: 8080
      initialDelaySeconds: 2
      timeoutSeconds: 2
      periodSeconds: 5
      successThreshold: 1
      failureThreshold: 3
    terminationMessagePath: /dev/termination-log
    name: filebrowser
    livenessProbe:
      httpGet:
        path: /health
        port: 8080
      initialDelaySeconds: 10
      timeoutSeconds: 2
      periodSeconds: 10
      successThreshold: 1
      failureThreshold: 3
    startupProbe:
      httpGet:
        path: /health
        port: 8080
      failureThreshold: 30
      periodSeconds: 10
    env:
      - name: TZ
        value: UTC
    ports:
      - name: http
        containerPort: 8080
        protocol: TCP
    imagePullPolicy: IfNotPresent
    securityContext:
      runAsNonRoot: true
      allowPrivilegeEscalation: false
      capabilities:
```


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```
              drop:
                - ALL
          volumeMounts:
            - name: config
              mountPath: /config
            - name: filebrowser-config
              mountPath: /.filebrowser.json
              subPath: .filebrowser.json
            - name: data
              mountPath: /files
          image: 'filebrowser/filebrowser:v2.18.0'
      automountServiceAccountToken: true
      serviceAccount: default
      volumes:
        - name: config
          persistentVolumeClaim:
            claimName: filebrowser-config
        - name: filebrowser-config
          configMap:
            name: my-filebrowser-config
            defaultMode: 420
        - name: data
          persistentVolumeClaim:
            claimName: files
      dnsPolicy: ClusterFirst
  strategy:
    type: RollingUpdate
    rollingUpdate:
      maxUnavailable: 1
      maxSurge: 1
  revisionHistoryLimit: 3
  progressDeadlineSeconds: 600

---
kind: Service
apiVersion: v1
metadata:
  name: filebrowser
  labels:
    app: filebrowser
    app.kubernetes.io/component: filebrowser
    app.kubernetes.io/instance: filebrowser
```

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```
    app.kubernetes.io/name: filebrowser
    app.kubernetes.io/part-of: filebrowser-app
spec:
  ports:
    - name: 8080-tcp
      protocol: TCP
      port: 8080
      targetPort: 8080
  type: ClusterIP
  selector:
    app: filebrowser

---
kind: Route
apiVersion: route.openshift.io/v1
metadata:
  name: filebrowser
  labels:
    app: filebrowser
    app.kubernetes.io/component: filebrowser
    app.kubernetes.io/instance: filebrowser
    app.kubernetes.io/name: filebrowser
    app.kubernetes.io/part-of: filebrowser-app
  annotations:
    openshift.io/host.generated: 'true'
spec:
  to:
    kind: Service
    name: filebrowser
    weight: 100
  port:
    targetPort: 8080-tcp
  tls:
    termination: edge
    insecureEdgeTerminationPolicy: Redirect
  wildcardPolicy: None
```

Verify the Application

From the CLI:

```
oc get pods,pvc -n filebrowser
```


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All pods should be `Running` and PVCs `Bound` within a minute or two.

From the OpenShift Web Console:

Navigate to Networking › Routes (project: `filebrowser`) and click the **Location** URL to open the filebrowser in a new tab.

Log In and Add Test Data

1. Log in with **Username:** `admin` / **Password:** `admin`.
2. Use New folder and Upload to create a couple of directories and upload a few files — these will be verified after restore.

Assign Backup Policies

There are two ways to assign policies in the **Fusion console**. Either method works for this lab.

Method A: Via the Applications Menu

1. In the Fusion Console, navigate to Applications.
2. Select the `filebrowser` checkbox and click Assign backup policy.
3. Select both `daily-snapshot` and `weekly-backup` checkboxes.
4. Leave **Back up now** enabled (green toggle). Click Save.

Method B: Via Backed Up Applications

1. Navigate to Backup & restore › Backed up applications.
2. Click Protect apps +.
3. Select the local cluster from the drop-down. Select `filebrowser` from the list and click Next.
4. Select both `daily-snapshot` and `weekly-backup`. Leave **Back up now** enabled. Click Assign.

A confirmation notification appears in the upper-right corner when policies are successfully assigned.

Monitor Backup Progress

1. Navigate to Backup & restore › Backed up applications and click `filebrowser`.
2. Click the **Backups** tab to track real-time progress.
3. The status transitions: `Pending` → `In progress` → `Completed`.

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From the CLI watch backup-related pods in the fusion namespace:

```
oc get pods -n ibm-spectrum-fusion-ns -w
```

Restore to a New Namespace

Run the Restore

1. From the **Backed up applications** page, click `filebrowser` to open its overview page.
2. Click Restore.
3. Set **Target cluster** to `This cluster`.
4. Select **Create a new project** and enter `filebrowser2`. Click Next.
5. Select the most recent completed backup from the list. Click Next.
6. Review the **Before restore / After restore** summary and click Restore.
7. Confirm with Restore in the dialog.

Verify the Restore

Monitor restore progress from Backup & restore › Jobs › Restore tab or from the CLI watch pods coming up in the restored namespace:

```
oc get pods,pvc -n filebrowser2 -w
```

Once the job shows **Completed**:

1. On the OpenShift cluster UI navigate to Networking › Routes (project: `filebrowser2`) and click the **Location** URL.
2. Log in with **Username:** `admin` / **Password:** `admin`.
3. Confirm the directories and files uploaded before the backup are present — the restore is verified.

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Backup Consistency Concepts

Introduction

When IBM Fusion backs up an application that has multiple persistent volumes, the method used to coordinate those snapshots determines how reliably the application can be restored. Understanding the three levels of backup consistency helps you choose the right protection strategy for each workload.

No Consistency

With no consistency, snapshots of individual persistent volumes are taken sequentially — one after another — rather than at the same point in time. If an application is actively writing data during this window, some volumes will reflect state from before a write while others reflect state from after it. The resulting backup is internally inconsistent and may not restore to a working application state.

Crash Consistency

IBM Fusion's default backup behavior is **crash consistent**. All persistent volume snapshots for an application are triggered simultaneously, so they all capture data at exactly the same instant. This is equivalent to what would happen if power were suddenly cut to the cluster — all writes stopped at the same moment, so all volumes are consistent with each other.

Crash consistency is sufficient for many stateless or lightly stateful applications. However, applications that maintain in-memory buffers — such as databases — may have uncommitted transactions or partially written data in those buffers at the moment of snapshot. The snapshot captures the on-disk state only, not what is in memory.

Application Consistency

Application-consistent backups go a step further. Before the snapshot is taken, the application is instructed to:

1. Finish any in-flight transactions
2. Flush all in-memory buffers to disk
3. Enter a quiesced (paused write) state

The snapshot is then taken while the application is quiesced, capturing a fully coherent on-disk state.

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Afterward, the application is instructed to resume normal operation.

This is the same approach used when quiescing a database for a hot backup — for example, calling `db.fsyncLock()` on MongoDB or `pg_start_backup()` on PostgreSQL.

Application-consistent backups are critical for:

- **Databases** — MongoDB, PostgreSQL, MySQL, Db2, etc.
- **Message queues and streaming platforms** — Kafka, RabbitMQ
- **Any application where uncommitted in-memory state must not be lost**

Fusion Recipes

IBM Fusion implements application-consistent backups through **Recipes** — Kubernetes Custom Resources (`kind: Recipe`) that define a custom backup and restore workflow for a specific application type.

A Recipe has three specification elements:

Groups

Define sets of resources or PVCs that are processed together. A group can include or exclude specific resource types (e.g., exclude events, pods, replicasets).

Hooks

Execute commands inside application pods before and after snapshots — for example, putting MongoDB into hot backup mode, or waiting until all expected replicas are Ready before restoring.

Workflows

Define the ordered sequence of groups and hooks for both the backup and restore operations.

IBM maintains a public library of recipes for common applications and databases in the IBM Storage Fusion GitHub repository. The MongoDB recipe used in the next lab is maintained there and referenced directly.

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Application-Consistent Backups with Recipes

Introduction

In this lab you will deploy the Pacman application backed by a MongoDB database, apply an IBM Fusion Recipe for application-consistent backups, trigger a backup, and verify that the MongoDB `fsyncLock` and `fsyncUnlock` hooks ran during the backup sequence. You will then configure Backup Service Protection to safeguard the Backup & Restore control plane itself.

Note This lab requires an active IBM Fusion reservation on IBM Technology Zone (TechZone). For step-by-step guidance on creating an IBMid, reserving an environment, and retrieving your cluster credentials, see the TechZone Quick Start Guide.

Note Review Backup Consistency Concepts before proceeding if you are not familiar with crash consistency vs. application consistency.

Deploy the Pacman Application

Create the Project

From the OpenShift Web Console:

1. Navigate to Home › Projects. Click Create Project.
2. Enter `pacman` in the **Name** field and click Create.

From the CLI:

```
oc new-project pacman
```

Import the Application YAML

1. Verify the active project is `pacman`.
2. Click the Quick create icon (+) in the masthead and select **Import YAML**.
3. Paste the following YAML and click Create:

```
kind: PersistentVolumeClaim
apiVersion: v1
```

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```
metadata:
  name: mongo-storage
spec:
  storageClassName: ocs-storagecluster-ceph-rbd
  accessModes:
    - ReadWriteOnce
  resources:
    requests:
      storage: 8Gi

---

apiVersion: apps/v1
kind: Deployment
metadata:
  labels:
    name: mongodb
    app: mongodb
  name: mongodb
spec:
  replicas: 1
  selector:
    matchLabels:
      name: mongodb
  template:
    metadata:
      labels:
        name: mongodb
        app: mongodb
    spec:
      containers:
      - image: docker.io/library/mongo:5.0 # was: bitnami/mongodb:5.0.24-debian-11-
r20
        name: mongodb
        ports:
        - name: mongodb
          containerPort: 27017
        volumeMounts:
          - name: mongo-db
            mountPath: /data/db # was: /bitnami/mongodb/data/db
      volumes:
        - name: mongo-db
```


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```
          persistentVolumeClaim:
            claimName: mongo-storage

---

apiVersion: v1
kind: Service
metadata:
  labels:
    name: mongodb
  name: mongo
spec:
  type: ClusterIP
  ports:
    - port: 27017
      targetPort: 27017
  selector:
    name: mongodb

---

apiVersion: apps/v1
kind: Deployment
metadata:
  labels:
    name: pacman
  name: pacman
spec:
  replicas: 1
  selector:
    matchLabels:
      name: pacman
  template:
    metadata:
      labels:
        name: pacman
    spec:
      containers:
      - image: quay.io/jpacker/nodejs-pacman-app:latest
        name: pacman
        ports:
        - containerPort: 8080
```

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```
          name: http-server
      initContainers:
      - name: db-check
        image: busybox:latest
        command: ['sh', '-c', 'echo -e "Checking for the availability of MongoDB
Server deployment"; while ! nc -z mongo 27017; do sleep 1; printf "-"; done; echo -e
"  >> MongoDB Server has started";']

---

apiVersion: v1
kind: Service
metadata:
  name: pacman
  labels:
    name: pacman
spec:
  type: ClusterIP
  ports:
    - port: 80
      targetPort: 8080
      protocol: TCP
  selector:
    name: pacman

---

apiVersion: route.openshift.io/v1
kind: Route
metadata:
  name: pacman
  labels:
    app.kubernetes.io/name: pacman
spec:
  path: "/"
  to:
    kind: Service
    name: pacman
  tls:
    termination: edge
    insecureEdgeTerminationPolicy: Redirect
  wildcardPolicy: None
```


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Verify and Play

1. Navigate to Networking › Routes (project: `pacman`) and click the **Location** URL.

   Note It may take a few minutes for the Pacman application to become fully online.

2. Play a round of Pacman. When the Game Over screen appears, enter your name and click Save.
3. Click View Highscore List to confirm the score was written to the MongoDB database. These scores will be verified after restore.

Apply the MongoDB Backup Recipe

The MongoDB recipe is maintained in the IBM Storage Fusion public GitHub repository. It is referenced directly rather than stored locally because it is owned and versioned by the upstream project.

1. Navigate to mongodb-image-based-backup-restore.yaml in GitHub. Click Copy raw file to copy the contents.
2. Click the Quick create icon (+) in the masthead and select **Import YAML**.
3. Paste the recipe content and click Create.

The recipe defines:

- **Groups** — all volumes and resources in the `pacman` namespace, excluding events, pods, and replicasets.
- **Hooks** —
  (1) Validate replica count is `Ready`
  (2) Put MongoDB into and out of hot backup mode (`fsyncLock` / `fsyncUnlock`).
- **Workflows** — backup sequence: back up resources → `fsyncLock` → snapshot volumes → `fsyncUnlock`. Restore sequence: create volumes first, then restore resources, wait for replicas Ready.

Assign the Policy and Attach the Recipe

Assign the weekly-backup Policy

1. Navigate to Applications in the Fusion Console.
2. Select the `pacman` checkbox and click Assign backup policy.

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3. Select `weekly-backup`.
4. **Set the Back up now toggle to disabled** (no green checkmark).

> **Important** Do not enable **Back up now** here. Pacman does not generate sufficient database activity to safely trigger an initial backup without the recipe attached. An initial backup without the recipe may cause restore failures due to Change Block Tracking (CBT).

5. Click Save.

Attach the Recipe to the PolicyAssignment

First, find the exact name of the PolicyAssignment created for the pacman application. The naming format is `<app>-<policy>-<cluster-name>`.

Option 1: From the CLI:

```
oc get policyassignment -n ibm-spectrum-fusion-ns | grep pacman
```

Note the full name (for example: `pacman-weekly-backup-apps.xxxx.ocp.techzone.ibm.com`), then patch it to attach the recipe:

```
oc -n ibm-spectrum-fusion-ns patch policyassignment <POLICY-ASSIGNMENT-NAME> \
  --type merge \
  -p '{"spec":{"recipe":{"name":"mongodb-image-based-backup-restore-
recipe","namespace":"ibm-spectrum-fusion-ns","apiVersion":"spp-data-
protection.isf.ibm.com/v1alpha1"}}}'
```

Option 2: From the OpenShift Web Console:

1. Navigate to Home › Search and switch to the `ibm-spectrum-fusion-ns` project.
2. In the **Resources** drop-down, search for `PolicyAssignment` and select it.
3. Click the PolicyAssignment for the pacman application.
4. Click the **YAML** tab.
5. Add the following four lines under `spec:`, after `runNow:`, then click Save:

```
recipe:
  apiVersion: spp-data-protection.isf.ibm.com/v1alpha1
  name: mongodb-image-based-backup-restore-recipe
  namespace: ibm-spectrum-fusion-ns
```


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Run the Backup and Verify Hooks

Trigger the Backup

1. Navigate to Applications in the Fusion Console.
2. Select the `pacman` checkbox and click Back up now.
3. Confirm with Back up.

Verify Recipe Hooks in the Job Log

1. Navigate to Backup & restore › Jobs.
2. Click the `pacman-weekly-backup-<cluster-name>` job.
3. Expand the **Backup sequence** section — confirm that the MongoDB hook steps (`fsyncLock`, `fsyncUnlock`) appear in the pipeline.
4. Click **Log view** for additional detail.

Verify Hooks in the MongoDB Pod Logs

From the OpenShift Web Console:

1. Navigate to Workloads › Pods (project: `pacman`).
2. Click the `mongodb-*` pod and select the **Logs** tab. Enable **Wrap lines**.
3. Scroll through the logs and confirm the `fsyncLock` command was executed during the backup.
4. Confirm the `fsyncUnlock` command appears shortly after, indicating MongoDB was successfully taken out of hot backup mode.

From the CLI:

```
oc logs -n pacman -l app=mongodb --tail=100 | grep -E 'fsyncLock|fsyncUnlock'
```

Both `fsyncLock` and `fsyncUnlock` entries should appear in the output confirming the recipe hooks ran correctly.

Configure Backup Service Protection

Backup Service Protection backs up the Backup & Restore control plane itself to an S3 bucket. If the Hub cluster fails, you can use this backup to restore the service on a new cluster.

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Note Service protection covers only the Backup & Restore service on the Hub cluster. It does not protect other Fusion services, OCP cluster state, Data Foundation, or disaster recovery configuration.

1. Navigate to Backup & restore › Service protection.
2. Click the **Configure service backups** tile.
3. Click the **S3 Compliant** tile and click Next.
4. Enter the S3 endpoint and credentials from a new OBC or an existing accessible bucket:
   - **Location name:** `service-protection-location`
   - **Endpoint**, **Bucket**, **Access Key**, **Secret Key** from your OBC
5. Click Add. A notification confirms the location is being added.
6. Click Define schedule.
7. Configure the schedule:
   - **Frequency:** `Weekly`
   - **Schedule:** `Sunday`
   - Set a **Start time**, **End time**, and **Timezone**.
8. Click Create policy. Leave **Initiate Service backup now** selected.
9. Confirm that the Service protection page shows the schedule and that an initial backup has started.

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Spoke Cluster Restore

Introduction

In a Hub-and-Spoke topology, a single Fusion Hub cluster manages backups for applications running on multiple Spoke clusters. In this lab you will connect a second TechZone cluster as a Spoke, back up an application on the Hub using the `object1` location created in Locations & Policies, and restore it to the Spoke.

Note This lab requires an active IBM Fusion reservation on IBM Technology Zone (TechZone). For step-by-step guidance on creating an IBMid, reserving an environment, and retrieving your cluster credentials, see the TechZone Quick Start Guide.

Important This lab requires two TechZone IBM Fusion environments. The cluster with the full Backup & Restore service installed is the **Hub**. The second cluster is the **Spoke**. Keep both browser sessions open — you will switch between them throughout this lab.

Install the Backup & Restore Agent on the Spoke

Step 1: Generate the Connection Snippet on the Hub

1. In the Fusion Console on the **Hub** cluster, navigate to Backup & restore › Topology.
2. Click Connect cluster.
3. Click the **Use Fusion UI** tile.
4. Click Copy snippet to copy the connection snippet to your clipboard.

Step 2: Install the Agent on the Spoke

1. In the Fusion Console on the **Spoke** cluster, navigate to Services.
2. Click the **Backup & Restore Agent** tile.
3. Click Install.
4. Paste the connection snippet from Step 1 into the **Hub Connection Snippet** field.
5. Open the **Storage class** drop-down and select `ocs-storagecluster-ceph-rbd`.

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6. Click Install.
7. Wait for the **Installing...** progress bar to complete and the service to report **Healthy**.

Step 3: Verify the Connection on the Hub

1. Return to the Fusion Console on the **Hub** cluster.
2. Navigate to Backup & restore › Topology.
3. Confirm the Spoke cluster is listed as **Connected** and **Healthy**.

Verify the Backup Location is Accessible

The `object1` backup location created in Locations & Policies uses the FDF NooBaa/MCG S3 route, which is accessible from both clusters over the TechZone network. No additional OBC is required.

Confirm the location is still **Connected**:

1. In the Fusion Console on the **Hub**, navigate to Backup & restore › Locations.
2. Verify `object1` shows **Connected**.

Note

IBM Fusion 2.13 enforces TLS certificate validation for backup locations. If the `object1` location shows a certificate error, you may need to provide the NooBaa CA bundle. Retrieve it with:

```
oc get cm noobaa-ca-bundle -n openshift-storage -o jsonpath='{.data.ca-bundle\.crt}'
```

Paste the output into the **CA Certificate** field when editing the location in the Fusion Console.

Create the Hub-Spoke Backup Policy

A dedicated policy scoped to the `object1` location is used for hub-spoke operations. If you already have `weekly-backup` pointing to `object1`, you can reuse it and skip this section.

1. Navigate to Backup & restore › Policies.
2. Click Add policy +.
3. Set:

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- **Policy name:** `hubspoke-backup`
- **Frequency:** `Monthly`, **Day:** `1`

4. Select the **Object Storage** tile and select `object1`.
5. Specify a **Time window** and click Create policy.

Deploy and Back Up the Application

Deploy Filebrowser in the hubspoke Namespace

1. Create the project:

   From the CLI:

   ```
   oc new-project hubspoke
   ```

   Or from the console: navigate to Home › Projects and click Create Project.

2. Click the Quick create icon (+) in the masthead → **Import YAML**. Paste the following YAML and click Create:

   ```
   kind: PersistentVolumeClaim
   apiVersion: v1
   metadata:
     labels:
       app: filebrowser
       app.kubernetes.io/component: filebrowser
       app.kubernetes.io/instance: filebrowser
       app.kubernetes.io/name: filebrowser
       app.kubernetes.io/part-of: filebrowser-app
     name: files
   spec:
     storageClassName: ocs-storagecluster-ceph-rbd
     accessModes:
       - ReadWriteOnce
     resources:
       requests:
         storage: 10Gi

   ---
   kind: PersistentVolumeClaim
   apiVersion: v1
   ```

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```
metadata:
  labels:
    app: filebrowser
    app.kubernetes.io/component: filebrowser
    app.kubernetes.io/instance: filebrowser
    app.kubernetes.io/name: filebrowser
    app.kubernetes.io/part-of: filebrowser-app
  name: filebrowser-config
spec:
  storageClassName: ocs-storagecluster-ceph-rbd
  accessModes:
    - ReadWriteOnce
  resources:
    requests:
      storage: 1Gi

---
kind: ConfigMap
apiVersion: v1
metadata:
  name: my-filebrowser-config
  labels:
    app: filebrowser
    app.kubernetes.io/component: filebrowser
    app.kubernetes.io/instance: filebrowser
    app.kubernetes.io/name: filebrowser
    app.kubernetes.io/part-of: filebrowser-app
data:
  .filebrowser.json: |
    {
      "port": 8080,
      "baseURL": "",
      "address": "",
      "log": "stdout",
      "database": "/config/database.db",
      "root": "/files"
    }

---
kind: Deployment
apiVersion: apps/v1
metadata:
```


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```
  name: filebrowser
  labels:
    app: filebrowser
    deployment: filebrowser
spec:
  replicas: 1
  selector:
    matchLabels:
      app: filebrowser
  template:
    metadata:
      labels:
        app: filebrowser
        app.kubernetes.io/component: filebrowser
        app.kubernetes.io/instance: filebrowser
        app.kubernetes.io/name: filebrowser
        app.kubernetes.io/part-of: filebrowser-app
    spec:
      restartPolicy: Always
      serviceAccountName: default
      schedulerName: default-scheduler
      enableServiceLinks: true
      terminationGracePeriodSeconds: 30
      securityContext: {}
      containers:
        - resources:
            requests:
              cpu: "100m"
              memory: "128Mi"
            limits:
              cpu: "500m"
              memory: "512Mi"
          readinessProbe:
            httpGet:
              path: /health
              port: 8080
            initialDelaySeconds: 2
            timeoutSeconds: 2
            periodSeconds: 5
            successThreshold: 1
            failureThreshold: 3
          terminationMessagePath: /dev/termination-log
```

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```
      name: filebrowser
      livenessProbe:
        httpGet:
          path: /health
          port: 8080
        initialDelaySeconds: 10
        timeoutSeconds: 2
        periodSeconds: 10
        successThreshold: 1
        failureThreshold: 3
      startupProbe:
        httpGet:
          path: /health
          port: 8080
        failureThreshold: 30
        periodSeconds: 10
      env:
        - name: TZ
          value: UTC
      ports:
        - name: http
          containerPort: 8080
          protocol: TCP
      imagePullPolicy: IfNotPresent
      securityContext:
        runAsNonRoot: true
        allowPrivilegeEscalation: false
        capabilities:
          drop:
            - ALL
      volumeMounts:
        - name: config
          mountPath: /config
        - name: filebrowser-config
          mountPath: /.filebrowser.json
          subPath: .filebrowser.json
        - name: data
          mountPath: /files
      image: 'filebrowser/filebrowser:v2.18.0'
  automountServiceAccountToken: true
  serviceAccount: default
  volumes:
```


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```
        - name: config
          persistentVolumeClaim:
            claimName: filebrowser-config
        - name: filebrowser-config
          configMap:
            name: my-filebrowser-config
            defaultMode: 420
        - name: data
          persistentVolumeClaim:
            claimName: files
      dnsPolicy: ClusterFirst
  strategy:
    type: RollingUpdate
    rollingUpdate:
      maxUnavailable: 1
      maxSurge: 1
  revisionHistoryLimit: 3
  progressDeadlineSeconds: 600

---
kind: Service
apiVersion: v1
metadata:
  name: filebrowser
  labels:
    app: filebrowser
    app.kubernetes.io/component: filebrowser
    app.kubernetes.io/instance: filebrowser
    app.kubernetes.io/name: filebrowser
    app.kubernetes.io/part-of: filebrowser-app
spec:
  ports:
    - name: 8080-tcp
      protocol: TCP
      port: 8080
      targetPort: 8080
  type: ClusterIP
  selector:
    app: filebrowser

---
kind: Route
```

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```
apiVersion: route.openshift.io/v1
metadata:
  name: filebrowser
  labels:
    app: filebrowser
    app.kubernetes.io/component: filebrowser
    app.kubernetes.io/instance: filebrowser
    app.kubernetes.io/name: filebrowser
    app.kubernetes.io/part-of: filebrowser-app
  annotations:
    openshift.io/host.generated: 'true'
spec:
  to:
    kind: Service
    name: filebrowser
    weight: 100
  port:
    targetPort: 8080-tcp
  tls:
    termination: edge
    insecureEdgeTerminationPolicy: Redirect
  wildcardPolicy: None
```

3. Navigate to Networking › Routes (project: `hubspoke`) and open the filebrowser URL.
4. Log in with **Username:** `admin` / **Password:** `admin`.
5. Create some directories and upload a few test files — these will be verified on the Spoke after restore.

Assign the hubspoke-backup Policy

1. In the Fusion Console on the **Hub**, navigate to Applications.
2. Select the `hubspoke` checkbox and click Assign backup policy.
3. Select `hubspoke-backup`. Leave **Back up now** enabled. Click Save.
4. Monitor Backup & restore › Backed up applications until `hubspoke` shows **Completed** before proceeding.

Restore to the Spoke Cluster

1. In the Fusion Console on the **Hub**, navigate to Backup & restore › Backed up applications.
2. Click `hubspoke` to open its details page.

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3. Click Restore.
4. In the **Cluster destination** drop-down, select the Spoke cluster — the one that does **not** show the **This cluster** tag.
5. Select **Use same project** and click Next.
6. Select the most recent backup time and click Next.
7. Review the summary and click Restore. Confirm with Restore.

Monitor the Restore Job

Navigate to Backup & restore › Jobs › Restore tab on the **Hub**. Click the `restore-hubspoke-<date>` job and monitor until **Completed**.

From the CLI on the Hub:

```
oc get pods -n ibm-spectrum-fusion-ns -w
```

Verify on the Spoke Cluster

1. In the OpenShift Web Console on the **Spoke**, navigate to Workloads › Pods (project: `hubspoke`).
2. Wait for the pod to show `1/1 Ready`. If the route is not accessible, restart the pod:

   ```
   oc delete pod -l app=filebrowser -n hubspoke
   ```

   Note This is a known post-restore behaviour — the application starts correctly but the route may not respond until the pod is restarted.

3. Navigate to Networking › Routes (project: `hubspoke`) on the **Spoke** and open the filebrowser URL.
4. Log in with **Username:** `admin` / **Password:** `admin`.
5. Confirm the directories and files created on the Hub are present — the cross-cluster restore is complete.

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